

Long-term agricultural diversification increases financial profitability, biodiversity, and ecosystem services: a second-order meta-analysis

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Sustainable agriculture in the 21st century requires the production of sufficient food, while reducing environmental impacts and safeguarding human livelihoods. Many studies have confirmed agricultural diversification with practices such as intercropping, organic farming, and soil inoculations, as suitable pathways to achieve these goals, but the long-term viability of socioeconomic and ecological benefits is uncertain. Here, we use a second-order meta-analysis to quantify long-term effects of agricultural diversification practices on socioeconomic and ecological benefits based on 120 years of data from 184 meta-analyses with 6741 effect sizes. Diversification increases financial profitability, biodiversity, pollination, soil quality, and carbon sequestration from 37 to 189% over 20 years, and we find no significant effects on crop yield, pest control, and climate regulation. Non-crop and crop diversification practices and use of organic amendments increase or maintain most of these benefits over time. Trade-off analysis between yield and all services shows win-win outcomes during the first 40 years. Our synthesis provides the urgently needed evidence for farmers and other decision-makers that diversification increases long-term financial profitability, biodiversity, soil quality, and climate change mitigation benefits. A large-scale implementation of agricultural diversification requires, however, careful consideration of factors that mediate these benefits. Then, agricultural diversification can be upscaled for long-term socioeconomic and nature-positive outcomes and, ultimately, for a global food system transformation.

Transforming the global food system requires nature-positive production systems^{1,2} using agroecological farming methods based on agricultural diversification practices to achieve sustainable food production^{3–5}. Agricultural diversification is based on farming practices, such as crop rotation, agroforestry, or organic farming, that

purposely integrate functional diversity into production systems to enhance or maintain the services essential to crop production⁶. Several syntheses have confirmed the socioeconomic (i.e., yield and profitability) and environmental (i.e., biological communities, soil quality, and climate mitigation) benefits of agricultural diversification

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practices for crop production worldwide^{7–9}. In Asia, for example, rice crop diversification practices reduce pesticide use and increase bio-control and crop yields¹⁰. These benefits have incentivized stakeholder groups to implement agricultural diversification across different countries. In China, agricultural diversification was recently promoted into the country's major agricultural policies¹¹. In Europe and the USA, several agricultural diversification practices, such as organic farming and crop/non-crop diversification, are advocated in agricultural policies for biodiversity conservation and climate change mitigation^{12–14}. Despite these successes, the long-term effects of agricultural diversification on socioeconomic and environmental factors remain unquantified and impede large-scale implementation by risk-averse farmers and promotion through policy makers.

Stakeholder groups require long-term evidence that diversified agricultural production can maintain and ideally improve livelihoods, economy, and the environment to mitigate climate change and external stressors^{15–17}. For instance, in staple foods like rice, the socioeconomic and environmental benefits of diversified rice farming are now well quantified¹⁸, but implementation success will depend on understanding temporal dynamics of such benefits¹⁵. In cash crops such as cocoa, smallholder farmers have made long-term investments in their tree plantations and may be reluctant to transition towards diversified agroforestry systems without understanding the long-term effects on profitability and yields¹⁹. More generally, global food security requires policy decisions that assure reaching the Sustainable Development Goals now more than ever^{20–22}. Moving towards an understanding and quantification of the long-term effects of agricultural diversification practices on socioeconomic and environmental factors - and the trade-offs between them is, hence, essential to the successful implementation of a global food system transformation^{20,23,24}.

Here, we conduct a global synthesis of meta-analyses to quantify the effects of agricultural diversification practices on socioeconomic and environmental factors for up to 120 years. Based on data from 184 meta-analyses with 6741 effect sizes based on 17,989 original studies (Supplementary Fig. 1), we quantify the influence of duration (years) of diversification practices on their effects on socioeconomic factors (crop yield and financial profitability), biological communities (biodiversity, pollination, and pest control), soil quality (water regulation, soil fertility, and nutrient cycling), and climate change mitigation (carbon sequestration and climate regulation). We also analyse the temporal trade-offs between crop yield and ecosystem services (Methods).

Results and discussion

Socioeconomic, biological community, soil quality, and climate change mitigation benefits mostly increase over time

Our second-order meta-regression models showed that agricultural diversification practices significantly increased benefits for six out of ten socioeconomic and environmental factors in the long term (Fig. 1, Supplementary Table 1). The analysis of socioeconomic factors showed that while agricultural diversification did not significantly change crop yield over time (Fig. 1), financial profitability increased significantly by 189% over 20 years ($\text{LnRR}=0.3633$) compared to short-term effects (<3 years, $\text{LnRR} = 0.1259$, Supplementary Fig. 2, Supplementary Table 2). Our analyses also showed that, over time, the probability of agricultural diversification having no effect on crop yield increased (Supplementary Fig. 1). These findings show that, while farmers may not experience significant benefits in terms of crop yield from diversified farming systems over time, their income improves, highlighting the socioeconomic viability of agricultural diversification in the long run³.

For biological communities, agricultural diversification significantly increased biodiversity by 64% over a 20-year period and pollination benefits by 629% over 10 years (Fig. 1, Supplementary

Table 2). These patterns provide evidence that agricultural diversification can play an important role not only in mitigating biodiversity losses from agriculture²⁵ but also in improving pollination services²⁶, which is highly important for global food security²⁷.

The mean effect sizes of soil quality and climate change mitigation, except for climate regulation, were significantly positive. Soil fertility, nutrient cycling and carbon sequestration had significant increases over time (Fig. 1), with benefits of 37% to 107% over a 20-year period (Supplementary Table 2). These patterns suggest that, in the long run, the benefits of agricultural diversification for soil fertility, nutrient cycling, and carbon sequestration are increasing, while the benefits of water regulation are maintained. The transformation from simple monocultures to diversified farming systems, therefore, can improve soil quality and carbon stocks in agricultural lands sustainably, with greater benefits in the long term^{28,29}.

Overall, our results provide evidence that agricultural diversification can help to achieve more sustainable and climate change resilient food production systems^{3,30} with financial profitability, biodiversity, pollination, soil fertility, nutrient cycling, and carbon sequestration being 22% to 110% higher after 20 years compared to non-temporal analysis results (Supplementary Fig. 2, Supplementary Table 2). Our analyses show that non-temporal analyses vastly underestimate the full potential that agricultural diversification can bring for socioeconomic and environmental factors after just two decades^{20,30}. These aspects support further the importance of considering the duration of agricultural diversification practices when evaluating their effects on the natural environment^{28,31,32}.

Our meta-regression analyses on the effects of six diversification practices on socioeconomic factors, biological community, soil quality, and climate change categories showed that almost half of the response variables became significantly positive or increased over time (green in Fig. 2, Supplementary Tables 3, 4). Non-crop diversification and organic amendments increased response variables in all four categories over time (Fig. 2, Supplementary Table 4), which provides realistic opportunities to address the twin challenges of biodiversity loss^{25,33} and climate mitigation^{34,35}. In addition, the long-term maintenance or significant improvement of soil fertility and nutrient cycling shows enhancement of soil quality^{34,36,37}. Long-term benefits of organic amendments highlight the potential of this practice to reduce chemical inputs, and hence, pollution and health repercussions^{38,39}. Therefore, these practices constitute the best options for shifting away from unsustainable and towards more biodiversity and nature-friendly farming practices^{3,20}. It is noteworthy, however, that certain practices, such as embedded natural habitats or flower strips require more land and may limit the yield per unit area. Additionally, the effectiveness of the uses of organic amendments depends on factors such as the application rate of organic inputs^{40,41}, as well as the availability of these inputs⁴². In some cases, the optimal quantity of organic materials required by certain farmlands for maximum benefits⁴³ might exceed the land's capacity to produce such material. In such cases, the use of external organic fertilizers may be necessary and could hinder large-scale implementation of these practices. Therefore, reaping the long-term benefits of these practices requires a nuanced consideration of implementation conditions and limitations.

Among all the types of agricultural diversification practices, crop diversification was the only practice that significantly increased financial profitability and resulted in non-significant effects or changes on yields over time (Fig. 2, Supplementary Table 4). Crop diversification can, hence, improve farmers' income over time⁴⁴ and ensure the long-term maintenance of crop yield⁴⁵. It is worth noting that, although crop diversification practices do not necessarily increase crop yield compared to conventional practices, they can contribute to food security through diet diversification and might help improve resistance to external shocks and climate change adaptation^{46–48}. These benefits may be limited by non-productive years in case of temporal

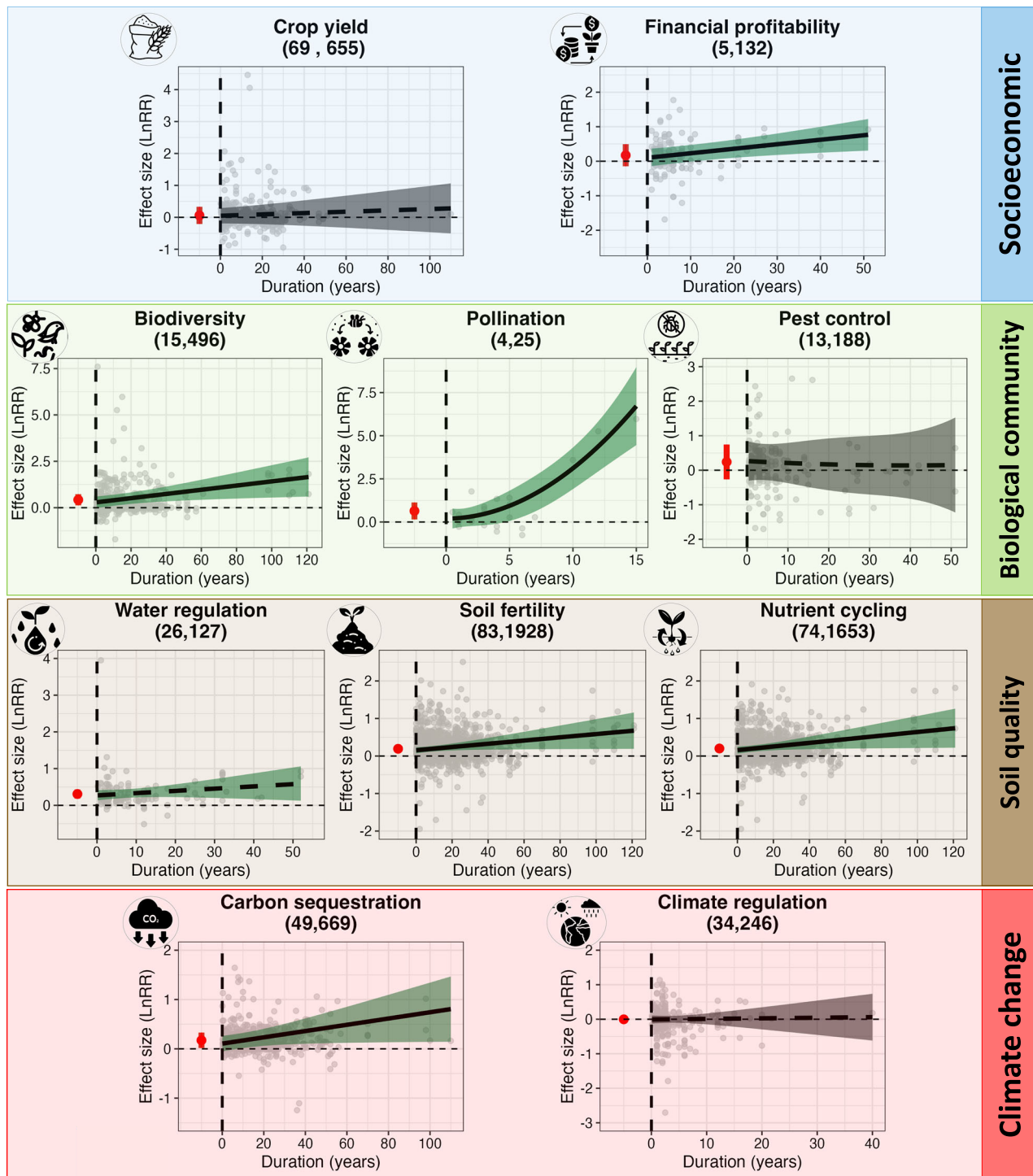


Fig. 1 | Effects of agricultural diversification on socioeconomic, biological community, soil quality, and climate change-related variables over time compared to non-temporal effects. Red point-range elements represent fitted values from non-temporal model predictions, with points indicating the estimates and vertical lines representing 95% confidence intervals (CIs). Black lines (solid or dashed) in the middle of ribbons represent fitted values from model predictions, with the ribbons indicating 95% confidence intervals (CIs). Respective solid or dashed lines inside ribbons represent significant and non-significant changes in effect sizes (LnRR) over time. Ribbon colours: grey = effects of diversification practices are non-significant; green = significant positive effect of diversification. In brackets are the number of studies and effect sizes. Icons used in this figure are

attributed as follows: crop yield by IconPai from the Noun Project under a CC BY 3.0 license; financial profitability by Slamblabs from Freepik under a CC BY 3.0 license; biodiversity by the Noun Project, no attribution required; pollination by Nithinan Tatah from the Noun Project under a CC BY 3.0 license; pest control by Koson Rattanaphan from the Noun Project under a CC BY 3.0 license; water regulation by Eucalypt from the Noun Project under a CC BY 3.0 license; soil fertility by Mindworld from the Noun Project under a CC BY 3.0 license; nutrient cycling by Yohoo from the Noun Project under a CC BY 3.0 license; carbon sequestration by Sorembe from the Noun Project under a CC BY 3.0 license; climate regulation by Karyative from the Noun Project under a CC BY 3.0 license.

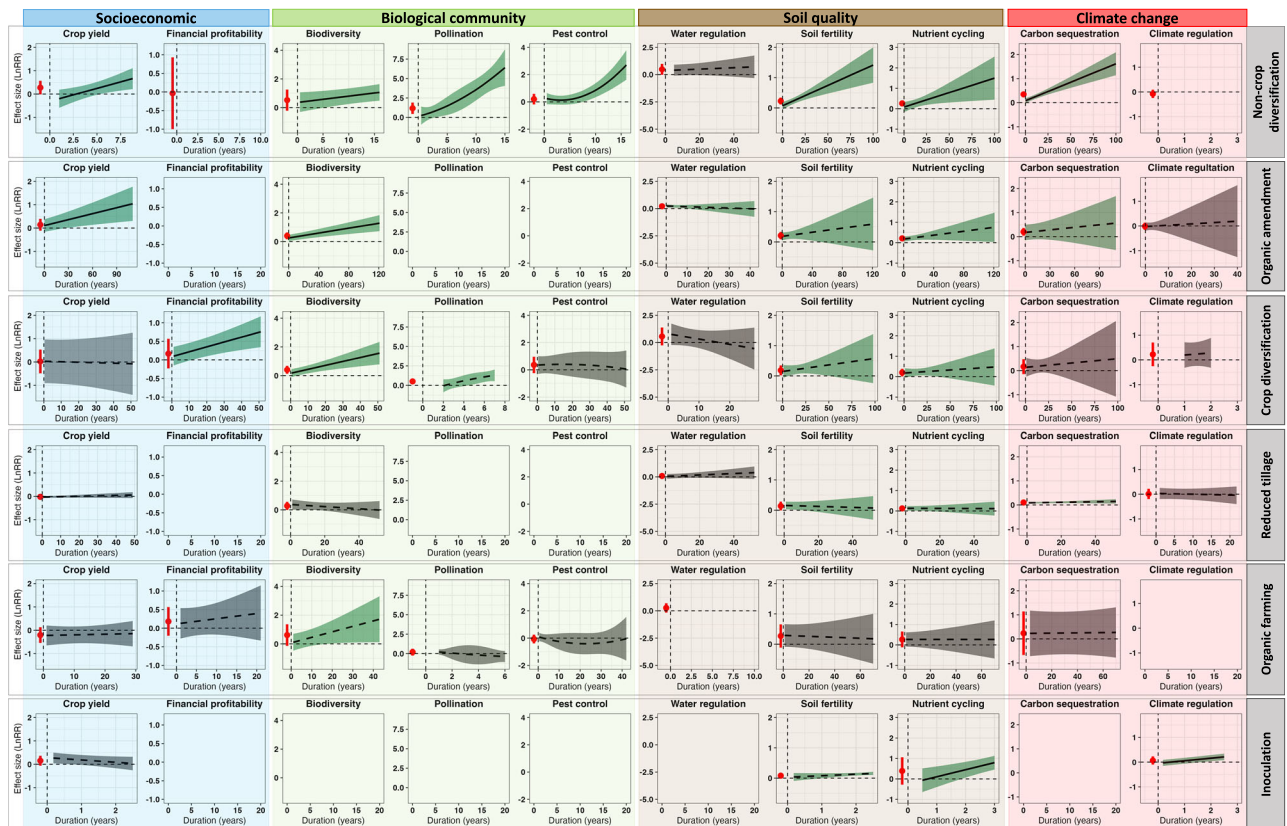


Fig. 2 | Effects of individual diversification practices on socioeconomic, biological community, soil quality, and climate change-related variables over time, compared to non-temporal effects. Red point-range elements represent fitted values from non-temporal model predictions, with points indicating the estimates and vertical lines representing 95% confidence intervals (CIs). Black lines (solid or dashed) in the middle of ribbons represent fitted values from model predictions, with the ribbons indicating 95% confidence intervals (CIs). Respective solid or

dashed lines inside ribbons represent significant and non-significant changes in effect sizes (LnRR) over time. Ribbon colours: grey = effects are non-significant over time; green = significant positive effect of diversification. For non-crop diversification, financial profitability and climate regulation predictions cover three years and one year, respectively. For organic farming, the water regulation predictions cover 33 years. Empty panels indicate research gaps due to data unavailability. See Supplementary Table 3 for the number of studies and effect sizes.

crop diversification and by land availability for spatial crop diversification⁴⁸. Our result also showed that crop diversification effects on pest control services did not change significantly over time, and there was no significant difference compared to conventional agriculture for up to 50 years (Fig. 2, Supplementary Table 4). This result indicates that some known benefits for instance of crop rotation on pest control may be counteracted⁴⁹, where environmental complexity increases incidence of some pests^{49–51}.

Reduced tillage showed slight yield increases over time but significant long-term benefits for carbon sequestration, while organic farming did not provide significant long-term benefits, except for biodiversity (Fig. 2, Supplementary Table 4). Overall, mainstreaming agricultural diversification and the combination of various practices may be a very effective and long-term strategy for global food system transformation^{5,9}. For instance, non-crop diversification strategies seem to hold most benefits as an individual practice, while organic farming shows an incremental increase in biodiversity benefits over time but has consistently lower yields (but see ref. 52). An effective combination of practices may help to transform some of the non-significant effects into positive long-term benefits^{8,9}.

Long term trade-offs between crop yield and other ecosystem services

We predicted trade-offs between crop yield and other services over time based on hierarchical regression models, in which all effect sizes were divided into loss and win outcomes (Methods). Overall, we detected a win-win situation between crop yield and all other services,

with a probability higher than 75% within the first 50 years of agricultural diversification practices, and then a lose yield-win ecosystem services was the most likely outcome (Fig. 3A; Supplementary Table 5). When examining the trade-off between yield and individual services, we found that increasing win-win situations with biodiversity, water regulation, and climate regulation were predominant (Fig. 3C, F and J, Supplementary Table 5). These results show that the positive and desirable outcomes of agricultural diversification practices not only remain but also become more probable over time².

Our analyses also showed win-win situations with financial profitability, soil fertility, nutrient cycling, and carbon sequestration for 7, 45, 47, and 25 years, respectively (Fig. 3), and lose yield-win ecosystem service situations becoming more common afterwards, interestingly including lose yield-win financial profitability. Our results indicate that compared to monocultures and conventional agriculture, agricultural diversification can improve crop yield in short terms while increasing farmers' income, improving soil nutrients, and capturing and storing carbon from the atmosphere in the long run, thus, achieving nature-positive systems and equitable livelihoods for the farmers^{1,53,54}. Government agri-environmental schemes should, hence, compensate for lower yields in diversified systems⁵⁵. It is important to note, however, that this is highly aggregated information and that these relationships over time will vary depending on location, crop type, and diversification practice.

Our results showed a predominance of lose yield-win pest control and pollination relationships over the years (up to 15 years for pest control, Fig. 3D, E), indicating that increases in pollination and pest

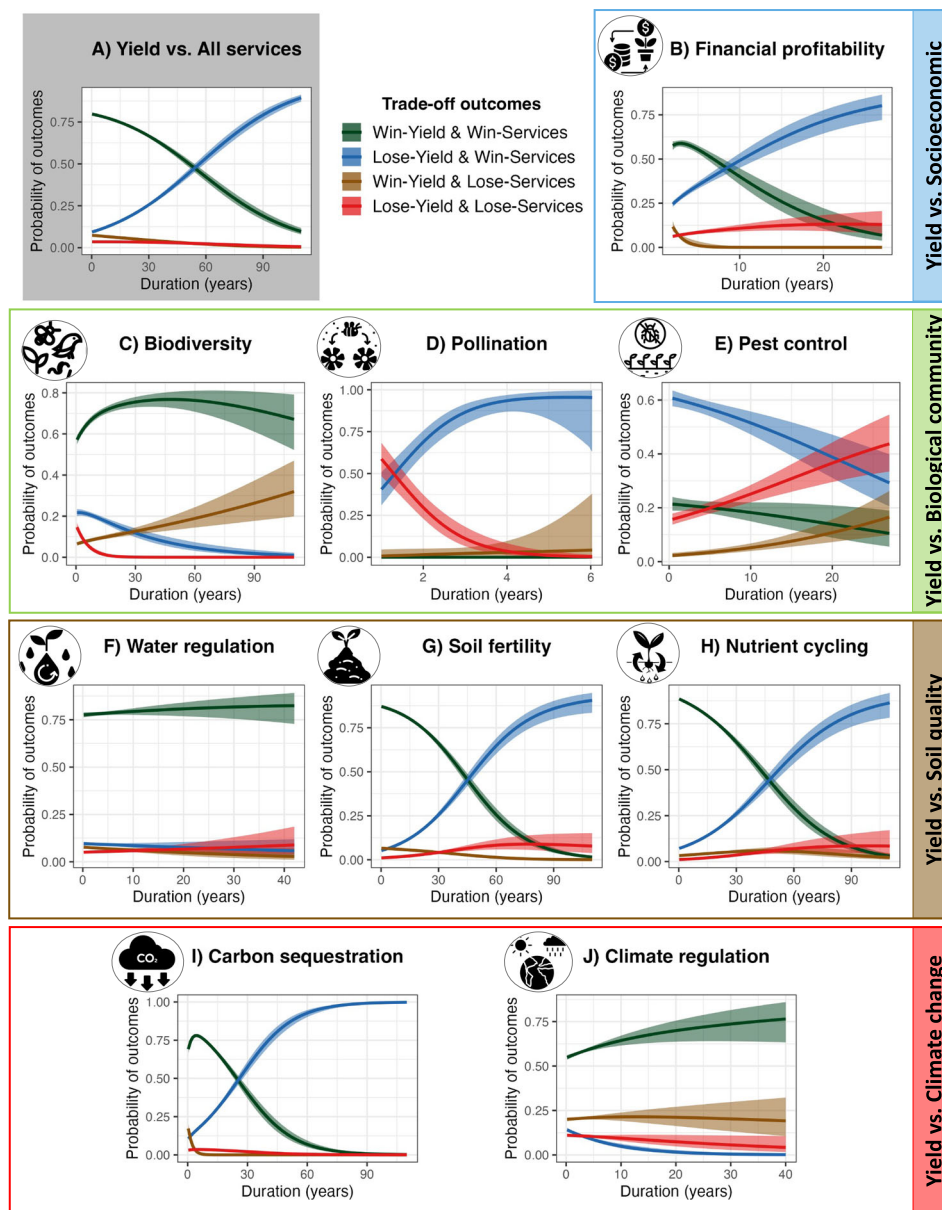


Fig. 3 | Trade-offs for agricultural diversification implementation over time: crop yield vs. Overall **A** and individual ecosystem services in the four categories of socioeconomic **B**, biological community **C–E**, soil quality **F–H**, and climate change (**I, J**). Solid lines surrounded by ribbons represent fitted values from model predictions, with the ribbons indicating 95% confidence intervals (CIs). Icons used in this figure are attributed as follows: crop yield by IconPai from the Noun Project under a CC BY 3.0 license; financial profitability by Slamlabs from Freepik under a CC BY 3.0 license; biodiversity from the Noun Project, no attribution required;

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control services do not always translate into higher crop yields. It is noteworthy, however, that while there is not necessarily a yield increase with pollination, 70% of all crops globally depend on natural pollination; hence, pollination is critical for yield stability^{27,56}. In addition, some studies have shown that pollination and pest control services can increase yield quality, thus improving the market value of crop yield⁵⁷.

Our analysis also revealed a temporal increase of the lose-lose relationship between crop yield and pest control services (Fig. 3E). However, the effects of diversification on pest control services were not statistically significant over time (Fig. 1, Supplementary Tables 1, 2), and the probability of this outcome increased over the years (Supplementary Fig. 1). This indicates that diversified farming may provide

concomitant benefits to some pest control agents and pests⁵⁸, which leads to some negative effects on crop yield. Use of agricultural technologies could help to reduce such undesirable effects of agricultural diversification via the use of pest-resistant varieties or more effective pest control technologies⁵⁹.

We also found an increase in lose-win relationship between crop yield and carbon sequestration over time (Fig. 3I). This indicates that with increasing effort for carbon capture, farmers face yield decrease, probably due to constraints associated with diversification practices such as smaller areas to plant crops in the case of non-crop diversification⁹. Thus, climate financing for diversified agriculture ought to consider such carbon-yield trade-offs and requires even more advocacy for diversification and agroecology at climate summit discussions.

Research gaps

Our work quantifies diversification effects for up to 120 years and allows us to identify existing temporal research gaps. First, short-term meta-analyses studies (0–10 years) on diversification effects on socioeconomic, biological community, soil quality, and climate change related variables were almost twice more common (146 studies and 3278 effect sizes) than long-term (>10 years) meta-analyses (83 studies and 2841 effect sizes). Likely, primary studies with long-term data are lacking, as funding opportunities for studies or experiments of >10 years are limited⁶⁰, but critically important to understand, for instance, diversification effects on pollination, climate regulation, and financial profitability (see Fig. 1 and Supplementary Fig. 3). Second, our trade-off analysis contained less data (47 studies, 2253 effect sizes) than the general meta-regressions (153 studies, 6119 effect sizes, Supplementary Figs. 3, 4). More studies are needed that concomitantly focus on diversification effects on yield or other services to enable analyses of multiple and individual diversification practices over time. Lastly, for individual diversification practices, we found no data on the effects of organic amendments and reduced tillage on financial profitability, pollination, and pest control over time (Fig. 2). Nevertheless, the use of these two practices is highly prevalent and advocated globally^{30,36}.

There were also no data on the effects of organic farming on climate regulation, and almost no effect sizes for long-term inoculation effects (see Fig. 2). These research gaps highlight the urgent need for more innovative funding mechanisms to support long-term studies on agricultural diversification, as well as quantitative syntheses of existing empirical research results. Such funding reform will ultimately help identifying better strategies to manage all trade-offs in diversified farming systems to maintain food security, alleviate poverty, and reduce environmental externalities of crop production^{61,62}.

The geographic distribution of primary studies represented in 123 out of 184 meta-analyses included in our synthesis revealed limited original studies from low and middle-income countries (Supplementary Fig. 5), highlighting a general geographic bias. For synthesis with long term effects of agricultural diversification (>20 years), the original studies from the majority of countries in Africa were underrepresented (Supplementary Fig. 6). This research gap highlights the necessity for more primary studies or at least synthesis of data from these regions, and more funding to explore the benefits of diversification practices in these contexts. Addressing this imbalance is critical to enable practitioners to inform decision making that can then help achieving the Sustainable Development Goals (SDGs)^{4,22}. Furthermore, research has demonstrated that geographic location significantly influences the outcomes of agricultural diversification practices^{47,63}, emphasizing the importance of region-specific studies to develop tailored and impactful strategies.

A long-term perspective to implement agricultural diversification

We show that non-temporal analyses have underestimated the benefits of agricultural diversification by 22 to 290%, particularly for financial profitability, biodiversity, and pollination (Supplementary Table 2). Our work quantifies the long-term benefits of agricultural diversification and can inform researchers, policymakers, and practitioners on urgently needed implementation strategies².

For the research community, it is important to focus on long-term projects that can disentangle the complex relationships between yield and various ecosystem services linked to biodiversity loss and climate change^{3,15}. Intuitively, a long-term focus will also address many remaining short-term research gaps on the way⁷. Clearly, addressing trade-offs between different ecosystem services requires multi-disciplinary collaborations and international partnerships such as the Agroecology Coalition, which can play a key role in rapidly advancing progress of the Sustainable Development Goals^{20,64}. Naturally, addressing long-term and applied research questions will require long-

term funding mechanisms from the public and private sectors⁶⁰. Such funding mechanisms will allow researchers to establish an integrated research network and databases such as LTSEr platforms⁶⁵ on a global scale, which will then promote open research, facilitating syntheses in social and ecological research⁶⁶.

For policymakers, our results provide critical evidence of how agricultural diversification can provide long-term support for biodiversity conservation, soil quality improvement, climate change mitigation, and at least maintain yields³⁰. While biodiversity benefits increase over time, there is a non-significant decline in crop yield despite the commonly expected trade-off associated with biodiversity integration in farmlands^{67–69}. The contribution of diversified farming systems to climate change mitigation increases over time, creating win-win solutions between crop yield and climate change regulation. Awareness and adoption of agricultural diversification are increasing in Europe and North America, but efforts need to be enhanced, particularly in Asia and Africa, to achieve global-scale targets of climate change mitigation and sustainable food production⁵⁵. Agri-environmental schemes and equivalent policies globally should compensate for higher yield gaps in diversified systems to alleviate trade-offs where they exist. Then, policies can help restore degraded and agricultural lands more effectively to reach climate and biodiversity targets⁷⁰.

Our work enables farmers to understand the short- and long-term benefits of agricultural diversification compared to conventional agriculture or monocultures. A core argument for its implementation is a sustainable production strategy that can maintain crop yields, increase financial profitability over time, and benefit the natural environment. It also enables broader landscape finance strategies, as our results provide detailed evidence of the anticipated trade-offs of agricultural diversification implementation and the associated risks⁶⁹. However, implementing diversification also requires careful consideration of the factors that mediate individual diversification practices beyond the binary comparison between monocultures and diversified systems, including geographic variations across scales, and climate change, all of which are critical to ensure the benefits of agricultural diversification practices^{71–74}. Thus, agricultural diversification provides further opportunities to shift towards a truly integrated strategy³³, whereby local culture, context, and practices are accounted for, and ecosystem services are fully utilized for diverse livelihoods beyond crop-related productions⁷⁵. In conclusion, our synthesis of the long-term benefits and trade-offs of agricultural diversification globally provides an urgently needed piece of the transformation puzzle for scientists, policymakers, and farmers to make informed and timely decisions for sustainable agricultural systems.

Methods

Literature search and data compilation

We systematically searched for peer-reviewed meta-analysis papers analysing the effects of agricultural diversification on socioeconomic, biological community, soil quality, and climate change using Web of Science (<https://www.webofscience.com/>) and Scopus (<https://www.scopus.com/>). For this, we followed the PRISMA review protocol (Supplementary Fig. 7). More specifically, we searched for all papers published until December 31, 2022, using keywords related to agricultural diversification practices and socioeconomic and ecological factors related to farming practices, and set meta-analysis as the type of study (for search strings, see Supplementary Table 6). Our research yielded 5008 papers, which we then screened using Rayyan (<https://www.rayyan.ai/>) to determine which papers met the criteria for full-text screening. We considered a paper relevant for full-text screening if it was about agricultural production systems and potentially diversified farming systems. In this study, we defined agricultural diversification or diversified farming systems as any practices that purposely incorporate functional diversity into farming systems⁶. During

screening, we also focused on crop yield, financial profitability, biodiversity, pollination, pest control, water regulation, soil fertility, nutrient cycling, carbon sequestration, and climate regulation, and if it was not a primary study. These criteria resulted in 3692 papers being excluded, leaving 1316 papers for full-text screening (Supplementary Fig. 7).

We screened the full text of 1316 papers individually and found 11 second-order meta-analyses papers related to the topic of this study that focused on at least one of our response variables (see Supplementary Table 7). We also screened the papers from those meta-analyses and ensured that all studies included in these studies were also included in the remaining papers for abstract screening. In each paper, we searched for the following information as inclusion criteria: (1) the study should be a meta-analysis with at least one effect size comparing diversified farming systems to non-diversified practices. (2) The effect size should measure at least one of the following related variables: crop yield, financial profitability, biodiversity, pollination, pest control, water regulation, soil fertility, nutrient cycling, carbon sequestration, and climate regulation (for more details on the variables, see Supplementary Table 8). (3) The number and list of original studies should be reported. (4) The study should also include information on the duration of diversification practices. In the case of multiple response variables from the same study, we ensured that each response was associated with an independent factor without the possibility of redundancy from one response to another and causing pseudo-replication of the underlying data and effects. Furthermore, some variables could be used as indicators of more than one category of response variables. In this case, we put the variable as a non-repeated response for the main or directly related response variable and as a repeated response for the other response variables. For instance, pollinator richness is considered in both pollination and biodiversity, but we put it in pollination as a non-repeated response and in biodiversity as a repeated response.

The above criteria narrowed the list of studies to 192 meta-analyses for data extraction (Supplementary Data 1). We extracted the mean and variance of effect sizes expressed in log response ratio (lnRR), response ratio (RR), and percentage of change and noted the qualitative effect (i.e., whether it was non-significant, significantly positive, or significantly negative). We used Plot Digitizer (<http://plotdigitizer.sourceforge.net/>) to extract data from the figures. We converted the effect sizes from RR and percentage of change into lnRR to obtain a uniform measurement. When the effect size was presented in standardized mean difference (SMD) such as *Hedges' d* or other effect sizes that could not be converted into lnRR without original data, we only extracted the trend (negative or positive) of the effect size as well as the statistical significance, that is, negative, non-significant, or positive. For each effect size, we noted the duration (years) of agricultural diversification practices, as well as the number of original comparisons. In cases where the number of replicates was not reported, we set the number of replicates to 1.

In this study, the duration represents the number of years during which agricultural diversification was practiced. When the duration of practice was provided as an interval in the meta-analysis results, we examined whether the authors provided the data with the exact duration of the practice associated with each effect size included in the analysis. When the data were provided, we calculated the effect sizes for all the durations provided using the same method described in the paper. When the data were not provided with exact duration or not provided at all, we considered the mean duration (e.g. for 5–10 years, the duration would become 7.5 years). When the duration was indicated as “>x” we added two years, as an educated guess since we did not find any suitable guidance from the literature, so >10 years would become 12 years to have a single duration for the corresponding effect size.

We categorised each response variable (effect size) into crop yield, financial profitability, biodiversity, pollination, pest control, water regulation, soil fertility, nutrient cycling, carbon sequestration, and climate regulation (see Supplementary Table 8). We also categorised agricultural diversification practices into six categories: non-crop diversification, crop diversification, organic amendment, reduced tillage, inoculation, and organic farming (Supplementary Table 9).

A strong synthesis should not repeat the original studies to guarantee that the effect sizes from all studies are independent. We avoided including meta-analyses with similar underlying data by first running an independence analysis of our 192 studies based on the references of their original studies. We calculated the percentage of original references shared by papers and identified 24 papers with more than 30% of shared original studies. We examined the focus and duration of the effect sizes in these studies. When two or more papers had the same response variable within the same duration, we retained the effect sizes from the most recent study. Using these criteria, we excluded eight papers, resulting in 184 meta-analysis papers with 6,741 effect sizes from 17,989 studies and 192,817 comparisons (Supplementary Fig. 7). We evaluated each study's quality based on the methods used and the reported results (for more details, see Supplementary Table 10), as these aspects determine the accuracy of the first-order meta-analysis results, which then affect the outcomes of second-order meta-analyses models⁶⁷. More specifically, we assigned a score of 1 or 2 to each of 8 different information from the paper based on their absence or presence, respectively. For each paper, the total score could vary between 8–16 and we considered a paper as low quality when the score was from 8 to 13 and high quality when from 14 to 16 (see supplementary Table 10).

Data analysis

We synthesized the effects of agricultural diversification on various response variables across all agricultural diversification practices combined and for each practice individually. We used second-order meta-analysis based on hierarchical meta-regression models, weighted with the number of comparisons behind each effect size to account for the number of observations behind each effect size.

For each model, we first performed null model comparisons to determine the best random effects structure for each category of the response variable. For this, we compared the AIC of null models with different random effects structures based on effect size IDs and study IDs (see Supplementary Table 11) to account for heterogeneity or between-study variations from the primary studies and effect sizes⁷⁶. Specifically, study ID as a random effect would, for instance, account for differences in how yield is measured in different crops under different management in different studies. Then, we ran models predicting the effects of diversification on crop yield, financial profitability, biodiversity, pollination, pest control, water regulation, soil fertility, nutrient cycling, carbon sequestration, and climate regulation individually, without a time moderator (hereafter moderator), to have reference values for non-temporal syntheses. We further ran the same models by adding the duration of diversification practices as a moderator for both overall and individual agricultural diversification practices. In addition, we performed another model selection based on AIC to determine whether the relationship between our response variable and durations of practice follows a linear, quadratic, cubic, or quartic relationship (see Supplementary Table 11), using the best random effects selected from null model selection. We used the *metafor* r-package⁷⁷ to run the meta-regression models, extracted the model results with the r-package *orchaRd*⁷⁸, and used *ggplot2* for visualization⁷⁹, in R version 4.4.3⁸⁰.

We conducted several tests to detect potential biases in our models. First, we used a funnel plot to visually detect the residuals of

the models against the inverse of the standard error (Supplementary Fig. 8) with a potential publication bias when the intercept significantly deviated from zero⁸¹. We ran Egger's regression tests to detect asymmetry of funnel plots (Supplementary Table 12), coupled with fail-safe N analyses based on the Rosenthal method (Supplementary Table 13). We tested the influence of the studies based on the standardized residuals and hat values (Supplementary Fig. 9), where the studies with high influence were those with twice the average hat value and with high standardized residuals⁸². Our tests did not reveal any significant issues associated with publication biases or outliers.

To ensure that our results were robust, we performed several sensitivity analyses to compare the outcomes. First, given that we included repeated effect sizes in the response variables, we ran a second-order meta-analysis using a dataset without any repeated responses. Second, since study quality might create some biases, we ran the same analysis, but this time with the data set of high-quality studies only. To obtain results from the most conservative analysis, we ran a third type of model with high-quality studies and without repeated responses. These models were run for both the overall effects of diversification and the duration of practices. Overall, the results from these different datasets did not show any significant differences from our main results (i.e., the full dataset; Supplementary Fig. 10). For carbon sequestration and nutrient cycling, excluding repeated variables from full dataset and from high-quality studies resulted in non-significant effects from 50 years, possibly because of data limitations, leading to a wider range of confidence intervals. It is worth noting that all estimates remained positive. Therefore, for the coherence of data analyses, we used the full dataset throughout.

We also performed several sensitivity analyses to check the robustness of our results. First, we determined whether including studies that report imprecise duration could change the outcome of our results. For this, we ran a separate model with only the effect sizes that had exact duration. When compared to the results of the full dataset, we found no significant differences between outcomes. Water regulation was the only exception for which the confidence interval overlaps with 0 from 30 years onwards (see Supplementary Fig. 11), possibly due to data limitations. It is worth noting that 95 % of the effect sizes included in the full dataset for the second-order meta-analysis had precise durations. Second, we ran an analysis excluding the single data point at 100 years for crop yield to assess its impact; third, we ran a separate analysis of the effects of diversification on soil fertility without soil pH. Both analyses showed no significant differences compared to the results from our main models (see supplementary Fig. 12).

We analysed the variation over time in the trade-offs associated with crop yield. We selected studies that tested diversification effects on crop yield and other services simultaneously and categorised each effect size as respective, win, or lose if the effect size was positive or negative, regardless of their statistical significance. This yielded four standard trade-off outcomes: win_yield - win_Service, win_yield - lose_Service, lose_yield -lose_Service, and lose_yield - win_Service. We then ran hierarchical regression models predicting these outcomes based on a multinomial distribution using the *nnnet* r-package⁸³, and we used the effect size of the crop yield paired with services, nested within the study ID, as random effects. We included the average of the original comparisons of crop yield and paired services as weights to account for the underlying number of comparisons. We ran the models for crop yield with all the effect size categories combined first, and then models with the individual categories of effect size categories with the duration of the practices as a moderator.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used in this study are part of a larger ongoing project and are currently being analyzed for additional publications. For this reason, public deposition of the complete dataset is not yet possible. However, a fully anonymized dataset, sufficient to verify the reproducibility of our methods and analyses, has been deposited in Open Science Framework repository and is accessible at <https://doi.org/10.17605/OSF.IO/ZPR6A>⁸⁴. The complete dataset can be made available from the corresponding authors upon request, subject to restrictions related to ongoing analyses and future publications. All summary statistics and aggregated results required to interpret and verify the findings are provided in the Supplementary Information.

Code availability

All R code used to conduct the analyses and generate the figures in this study has been deposited in the Open Science Framework (OSF) repository and is accessible at <https://doi.org/10.17605/OSF.IO/ZPR6A>⁸⁴.

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Author contributions

E.R. and T.C.W. developed the study concept. E.R. screened all papers, extracted the data, conducted all analyses, produced all figures, and wrote the original draft of the manuscript. T.C.W. secured funding. E. R. and T. C. W. edited and revised the manuscript and approved its submission.

Competing interests

The authors declare no competing interests.

Additional information

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